

Mitochondria and Energy

Oxidation of fatty acids (ACADM, CPT1A, CPT2), lactate transport (MCT1), respiratory chain and CoQ10 (NDUFB6, COQ2, NQO1), mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM), sirtuins, decouplers (UCP1-3), mitophagy (PINK1, FOXO3) and mitochondrial DNA

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Variants by theme	6
5. Mitochondrial DNA	9
6. Favourable	10
7. Unfavourable	11
8. Mechanisms of ageing	12
9. Practical suggestions	13
10. Limitations	14
11. Methods and sources	15
12. Annex: all variants analysed	16

1. Summary

What was seen. 27 SNPs were read in 20 nuclear genes (22 of chip, 5 imputed) and 4 polygenic scores, plus classical mutations of mitochondrial DNA.

Solid basis. No mitochondrial metabolic disease: **ACADM** without K329E or the rare pathogenic variant, **CPT1A** without Pro479Leu, DNA without classical pathogenic mutations (MELAS, MERRF, LHON). Biogenesis has the best genotype: **PGC-1α Gly482Gly** (without Ser482) and without Met612. **NDUFB6** (complex I) without the lowest expression alleles, **NQO1 *1/*1** (normally reduced CoQ10), **TFAM** without Ser12Thr, **SIRT6** without the stop-gain and without T/T. Strong mitochondrial defense: **SOD2 Ala/Ala** and **FOXO3** with the longevity allele.

Light spots. **SIRT3 Val208Ile** (1 copy; lower efficiency of mitochondrial sirtuin), **UCP1 –3826G** (1 copy; less thermogenesis), **MCT1 Asp490** (1 copy; intermediate lactate transport), PPARA intron 7 C (1 copy; minus PPARα), **CPT2 Ile368** (1 copy; only relevant with pathogenic variants) and **PINK1 Thr521** (1 copy, imputed).

Typical genetic fitness: grip strength p52.4 / p51.3 / p29.5 and walking pace p57.5.

Reading. Mitochondrial machinery is genetically good and responds well to the right stimulus: PGC-1α Gly/Gly is the genotype that most benefits from aerobic training. The light points (SIRT3, UCP1) point to the same: exercise, moderate cold and regular night fasting are levers.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 27 SNPs: 22 of the chip, 5 imputed and 0 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 5 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **Polygenic scores (PGS):** 4 PGS Catalog scores calculated locally with 1,996,529 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Percentages in the reference population. There is no direct score of mitochondrial function in the catalog; grip strength and gait rhythm are the closest functional markers.

Strength and physical capacity

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Handgrip strength (score A)	PGS002143 (Am J Hum Genet 2021)	946,066 966,283 (weighted coverage 97%)	52.4	TYPICAL	typical
Handgrip strength (score B)	PGS003513 (Nature 2023)	972,627 979,739 (weighted coverage 99%)	51.3	TYPICAL	typical
Handgrip strength (score C)	PGS001927 (Am J Hum Genet 2021)	71,148 73,846 (weighted coverage 96%)	29.5	TYPICAL	typical
Rhythm of gait	PGS001075 (PLoS Genet 2022)	6,688 7,531 (weighted coverage 90%)	57.5	TYPICAL	typical

4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable; in the "no judgment", the allele has no sense of good or bad clarity.

Oxidation of fatty acids and lactate

Fatty acids enter mitochondria by CPT1A/CPT2 and are oxidized by dehydrogenases (ACADM for the medium chain). MCT1 transports lactate between muscle, heart and brain.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ACADM	rs77931234	Lys329Glu (K329E)	A/A	chip	G (0 copy/s)	favourable	The most common cause of MCAD deficiency (medium chain β -oxidation; recessive)	High
ACADM	rs121434274	pathogenic variant	G/G	chip	A (0 copy/s)	favourable	Rare pathogenic variant of MCAD deficiency	High
CPT1A	rs80356779	Pro479Leu	G/G	chip	A (0 copy/s)	favourable	Low activity variant of CPT1A (long fatty acids in mitochondria)	High
CPT1A	rs2229738	Ala275Thr	C/C	chip	T (0 copy/s)	favourable	Light effect variable	High
CPT2	rs1799821	Val368Ile	G/A	chip	A (1 copy/s)	unfavourable (light)	Ile368 integrates the thermolabile haplotype of CPT2 (cases with fever or prolonged fasting only in carriers of pathogenic variants)	High
SLC16A1 (MCT1)	rs1049434	Glu490Asp	A/T	chip	T (1 copy/s)	unfavourable (light)	The Asp490 allele reduces ~35-40% lactate transport; heterozygous = intermediate	Moderate

Respiratory chain and CoQ10

The respiratory chain produces ATP; complex I (NDUFB6) is the gateway. The CoQ10 carries electrons and the NQO1 keeps it reduced.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NDUFB6 (complex I)	rs540467	promoter	C/C	imputation DR2=0.97	T (0 copy/s)	favourable	Lower NDUFB6 expression in muscle and lower insulin sensitivity	Moderate
NDUFB6 (complex I)	rs629566	promoter (CPG)	T/T	imputation DR2=0.98	C (0 copy/s)	favourable	It creates a CpG methylation site and is associated with the decline of NDUFB6 expression with age	Moderate
COQ2	rs4693075	intronic	G/C	chip	C (1 copy/s)	without judgment	Associated with statin myopathy in small studies (only relevant with statin)	Low
NQO1	rs1800566	Pro187Ser (*2)	G/G	chip	A (0 copy/s)	favourable	*2: almost zero enzyme; NQO1 keeps CoQ10 in the reduced form (ubiquinol)	High

Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)

PGC-1 α is the master regulator of mitochondrial biogenesis (induced by exercise and cold); PPAR α activates fat oxidation; TFAM maintains mitochondrial DNA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PPARGC1A (PGC-1α)	rs8192678	Gly482Ser	C/C	chip	T (0 copy/s)	favourable	Ser482: less PGC-1α, less oxidative fibers, lower response to aerobic training and more 2 type diabetes	High
PPARGC1A (PGC-1α)	rs3736265	Thr612Met	G/G	chip	A (0 copy/s)	favourable	Met612: variant associated with insulin resistance	Moderate
PPARA	rs1800206	Leu162Val	C/C	chip	G (0 copy/s)	without judgment	Val162 changes the lipid response	High
PPARA	rs4253778	intron 7	G/C	chip	C (1 copy/s)	unfavourable (light)	The C allele gives less PPARα (power profile; G = resistance) and is associated with ventricular hypertrophy	Moderate
TFAM	rs1937	Ser12Thr	G/G	chip	C (0 copy/s)	favourable	Missense variant of mitochondrial DNA transcription factor	High

Sirtuins

Desacetylases that respond to the energy state: SIRT1 (core), SIRT3 (mitochondria) and SIRT6 (repair of DNA, telomeres).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SIRT1	rs7895833	promoter	A/A	chip	G (0 copy/s)	neutral	Minor allele with weak and inconsistent associations	Low
SIRT3	rs11246020	Val208Ile	T/C	imputation DR2=1.00	T (1 copy/s)	unfavourable (light)	Ile208 reduces the catalytic efficiency of SIRT3 (mitochondrial deacetylase) and is associated with metabolic syndrome	High
SIRT6	rs107251	intronic	T/C	chip	T (1 copy/s)	favourable (mild)	T/T associated with lower survival in the elderly	Low
SIRT6	rs117385980	stop-gain	C/C	imputation DR2=1.00	T (0 copy/s)	favourable	Stop-gain allele (loss of SIRT6 function)	High

Decouplers and thermogenesis (UCP1-3)

UCP dissipate the gradient of protons in heat (UCP1, brown adipose tissue) or reduce the formation of radicals (UCP2, UCP3).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
UCP1	rs1800592	-3826A>G (G = C in strand +)	T/C	chip	C (1 copy/s)	unfavourable (light)	-3826G: less UCP1 and less thermogenesis in brown adipose tissue	High
UCP1	rs45539933	Ala64Thr	C/C	chip	T (0 copy/s)	favourable	Missense variant of UCP1	High
UCP2	rs659366	-866G>A (A = T in strand +)	C/T	chip	T (1 copy/s)	without judgment	-866A: more UCP2 (less mitochondrial reactive species, but less insulin secretion)	High
UCP2	rs660339	Ala55Val	G/A	chip	A (1 copy/s)	without judgment	Val55: more decoupling in some studies	Moderate
UCP3	rs1800849	-55C>T (T = A in strand +)	G/G	chip	A (0 copy/s)	without judgment	-55T increases muscle UCP3	Moderate

Mitophagy and mitochondrial defense

PINK1 marks the damaged mitochondria for mitophagy; FOXO3 induces mitophagy and SOD2 removes mitochondrial superoxide.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PINK1	rs1043424	Asn521Thr	C/A	imputation DR2=0.98	C (1 copy/s)	unfavourable (light)	Missense variant of PINK1 (mitophagy)	Moderate-low
FOXO3	rs2802292	intronic	G/G	chip	G (2 copy/s)	favourable	Allele of longevity; FOXO3 induces PINK1, mitophagy, SOD2 and autophagy	High
SOD2 (MnSOD)	rs4880	Val16Ala (Ala = G)	G/G	chip	G (2 copy/s)	favourable	Ala16: MnSOD more active	High

5. Mitochondrial DNA

Position	Disease	Result
m.3243A>G	MELAS, diabetes and maternal inheritance deafness	Absent (reference)
m.8344A>G	MERRF	Absent (reference)
m.11778G>A	Leber's optic neuropathy (LHON)	Absent (reference)
m.14484T>C	Leber's optic neuropathy (LHON)	Absent (reference)
m.3438G>A (MT-ND1)	—	Present; 3.the codon position, probably synonymous and without effect

The chip reads only about 1 500 positions of the mitochondrial DNA and does not measure the heteroplasmy (the proportion of mutated copies) or the number of copies. The **MT-RNR1** (aminoglycosides ototoxicity) is not determinable in this chip.

6. Favourable

- **No mitochondrial disease or β -oxidation:** ACADM (K329E and pathogenic variant) and CPT1A (Pro479Leu, Ala275Thr) without variants; DNA mitochondrial without MELAS, MERRF or LHON.
- **PGC-1 α Gly482Gly** and without Thr612Met: the best genotype for mitochondrial biogenesis and response to aerobic training.
- **NDUFB6** (rs540467 C/C, rs629566 T/T; imputed): complex I without the lowest expression alleles.
- **NQO1 *1/*1** (CoQ10 kept reduced) and **TFAM** without Ser12Thr.
- **SIRT6:** without the stop-gain (imputed) and without the T/T genotype of rs107251.
- **SOD2 Ala/Ala** and **FOXO3** rs2802292 G/G (longevity, mitophagy).
- **UCP1** without Ala64Thr.

7. Unfavourable

- **SIRT3 Val208Ile** (rs11246020, 1 copy; imputed): lower efficiency of mitochondrial sirtuin.
- **UCP1 –3826G** (rs1800592, 1 copy): less thermogenesis in brown adipose tissue.
- **MCT1 Glu490Asp** (rs1049434, 1 copy): intermediate lactate transport.
- **PPARA intron 7 C** (rs4253778, 1 copy): a little less PPAR α .
- **CPT2 Val368Ile** (1 copy): component of the thermolabile haplotype (relevant only with pathogenic variants).
- **PINK1 Asn521Thr** (rs1043424, 1 copy; imputed).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Mitochondrial dysfunction	PGC-1α Gly/Gly, NDUFB6, ACADM, CPT1A and DNA mitochondrial without variants SIRT3 Val208Ile and PINK1 Thr521 heterozygous	favourable
Oxidative Phosphorylation Decline	Complex I (NDUFB6) without the lowest expression alleles and CpG that reduces expression with age	favourable
Mitochondrial Defective Quality Control	FOXO3 of longevity; PINK1 Asn521Thr (1 copy)	mixed
Changes in Mitochondrial Dynamics	PGC-1α Gly482Gly (biogenesis); TFAM without variant	favourable
Mutations in the DNA Mitochondrial	Without MELAS, MERRF or LHON; unmeasured heteroplasmy	favourable
Redox imbalance	SOD2 Ala/Ala and NQO1 1/1; UCP2 –866A (1 copy)	favourable
Thermogenesis decline	UCP1 –3826G (1 copy); UCP3 –55C/C	unfavourable (light)
Insulin resistance	PGC-1α Gly/Gly and NDUFB6 favourable; SIRT3 Ile208 and PPARA C (1 copy each)	mixed
Nutrient Sensing Deregulation	Sirtuins: SIRT3 Ile208 heterozygous; SIRT6 without stop-gain; Common SIRT1	neutral
Sarcopenia	Handgrip strength p52.4 / p51.3 / p29.5 (typical); Intermediate MCT1	neutral

9. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Regular aerobic training, including 2 zone and intervals** (3–5 times/week): PGC-1 α Gly/Gly responds very well; it is the strongest lever of mitochondrial biogenesis. Suggestion (not applied).
2. **Strength training** 2–3 times/week for muscle mass and strength. Suggestion (not applied).
3. **Moderate exposure to cold** (cold shower, fresh environment) to activate brown adipose tissue (UCP1 – 3826G). Suggestion (not applied).
4. **Night fasting of 12–14 h** and no late food: activates sirtuins (SIRT3 Ile208) and PPAR α . Suggestion (not applied).
5. **CoQ10** only if there is statin with muscle pain (COQ2 rs4693075 heterozygous); NQO1 *1/*1 keeps CoQ10 active. Suggestion (not applied).
6. **Creatine** (already in the stack): supports muscle and brain energy. Suggestion (not applied).
7. **Avoid very prolonged fasting with fever or extreme effort** is a generic precaution; CPT2 Ile368 alone does not justify restrictions. Suggestion (not applied).
8. **Omega-3 3.2–4 g/day of EPA+DHA** and regular sleep (biogenesis and nocturnal mitophagy). Suggestion (not applied).

10. Limitations

- Small and added effects; do not measure the real mitochondrial function (e.g. VO2max, lactate, muscle spectroscopy).
- Mitochondrial DNA is read in few positions; heteroplasmy and copy number are not measured on the chip. MT-RNR1 is not determinable.
- The **PINK1 rs1043424** went from A/A (previous inference by close neighbors, moderate-low confidence) to C/A in imputation (DR2 0.98); the imputation is read.
- The VNTR of SIRT3 (longevity) is not read in the chip.
- Conclusions only from genes; do not include blood analysis.

11. Methods and sources

- **Genes and SNPs:** 27 SNPs chosen from the previous analysis of mitochondria and energy (50 lines), keeping the functional variants and removing the SNPs without effect demonstrated. The classical mutations of the mitochondrial DNA were read directly from the chip.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local statistical imputation (public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect on strand +; in the strand genes – the nomenclature of the gene was converted (e.g. UCP1 – 3826G = C in strand +). 5 palindromic SNPs assumed in the strand +.
- **Polygenic scores:** PGS Catalog, calculated with the imputed genome; percentiles in the reference population.
- **Comparison with previous analysis (02-10-2026):** The 22 genotypes of the chip are equal; from those imputed, only PINK1 rs1043424 differs (see Limitations).
- **Context Literature:** Ek 2001 (PGC-1 α Gly482Ser), Hirschey 2011 (SIRT3), Ling 2007 (NDUFB6).

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Oxidation of fatty acids and lactate	ACADM	rs77931234	A/A	chip	G	0	strand +	without the risk allele
Oxidation of fatty acids and lactate	ACADM	rs121434274	G/G	chip	A	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
Oxidation of fatty acids and lactate	CPT1A	rs80356779	G/G	chip	A	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
Oxidation of fatty acids and lactate	CPT1A	rs2229738	C/C	chip	T	0	strand +	without the risk allele
Oxidation of fatty acids and lactate	CPT2	rs1799821	G/A	chip	A	1	strand +	1 copy of risk allele
Oxidation of fatty acids and lactate	SLC16A1 (MCT1)	rs1049434	A/T	chip	T	1	strand + (palindromic; assumed +)	1 copy of risk allele
Respiratory chain and CoQ10	NDUFB6 (complex I)	rs540467	C/C	imputation DR2=0.97	T	0	strand +	without the risk allele
Respiratory chain and CoQ10	NDUFB6 (complex I)	rs629566	T/T	imputation DR2=0.98	C	0	strand +	without the risk allele
Respiratory chain and CoQ10	COQ2	rs4693075	G/C	chip	C	1	strand + (palindromic; assumed +)	1 copy of effect allele
Respiratory chain and CoQ10	NQO1	rs1800566	G/G	chip	A	0	strand +	without the risk allele
Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)	PPARGC1A (PGC-1 α)	rs8192678	C/C	chip	T	0	strand +	without the risk allele
Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)	PPARGC1A (PGC-1 α)	rs3736265	G/G	chip	A	0	strand +	without the risk allele
Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)	PPARA	rs1800206	C/C	chip	G	0	strand + (palindromic; assumed +)	0 copy of the effect allele
Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)	PPARA	rs4253778	G/C	chip	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Mitochondrial biogenesis (PGC-1 α , PPAR α , TFAM)	TFAM	rs1937	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Sirtuins	SIRT1	rs7895833	A/A	chip	G	0	strand +	A/A: common genotype
Sirtuins	SIRT3	rs11246020	T/C	imputation DR2=1.00	T	1	strand +	1 copy of risk allele
Sirtuins	SIRT6	rs107251	T/C	chip	T	1	strand +	T/C: does not have the unfavourable T/T genotype
Sirtuins	SIRT6	rs117385980	C/C	imputation DR2=1.00	T	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Decouplers and thermogenesis (UCP1-3)	UCP1	rs1800592	T/C	chip	C	1	strand +	1 copy of risk allele
Decouplers and thermogenesis (UCP1-3)	UCP1	rs45539933	C/C	chip	T	0	strand +	without the risk allele
Decouplers and thermogenesis (UCP1-3)	UCP2	rs659366	C/T	chip	T	1	strand +	1 copy of effect allele
Decouplers and thermogenesis (UCP1-3)	UCP2	rs660339	G/A	chip	A	1	strand +	1 copy of effect allele
Decouplers and thermogenesis (UCP1-3)	UCP3	rs1800849	G/G	chip	A	0	strand +	0 copy of the effect allele
Mitophagy and mitochondrial defense	PINK1	rs1043424	C/A	imputation DR2=0.98	C	1	strand +	1 copy of risk allele
Mitophagy and mitochondrial defense	FOXO3	rs2802292	G/G	chip	G	2	strand +	2 copies of the protective allele
Mitophagy and mitochondrial defense	SOD2 (MnSOD)	rs4880	G/G	chip	G	2	strand +	Ala/Ala: MnSOD more active